

AXIOM MATH · OCTOBER 6, 2026

Large and small models for mathematical discovery



Based on Albilich · PatternMiner · CircuitPilot

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ai.math.uw.edu

Albilich

Proof generation

A proof harness with long-horizon context management and CAS tool calls.

Based on joint work with **Ting Gong** and **Yong Yang**.

arXiv preprint · 2026 · Code and research records

Proof generation

Aletheia

Generate, verify, and revise a solution; use literature and computation throughout.

Feng et al. (2026)

Danus

Parallel workers contribute checked claims to a shared fact graph.

Liu et al. (2026)

Rethlas + Archon

Informal reasoning and theorem retrieval, followed by formalization in Lean 4.

Ju et al. (2026)

ProofCouncil

Author–critic exchanges combine mathematical writing with computer algebra.

Schmitt et al. (2026)

EIGENWEIGHTS FOR ARITHMETIC HIRZEBRUCH PROPORTIONALITY

TONY FENG

2. DECLARATION OF AI USAGE

The core mathematical content of this paper was entirely generated

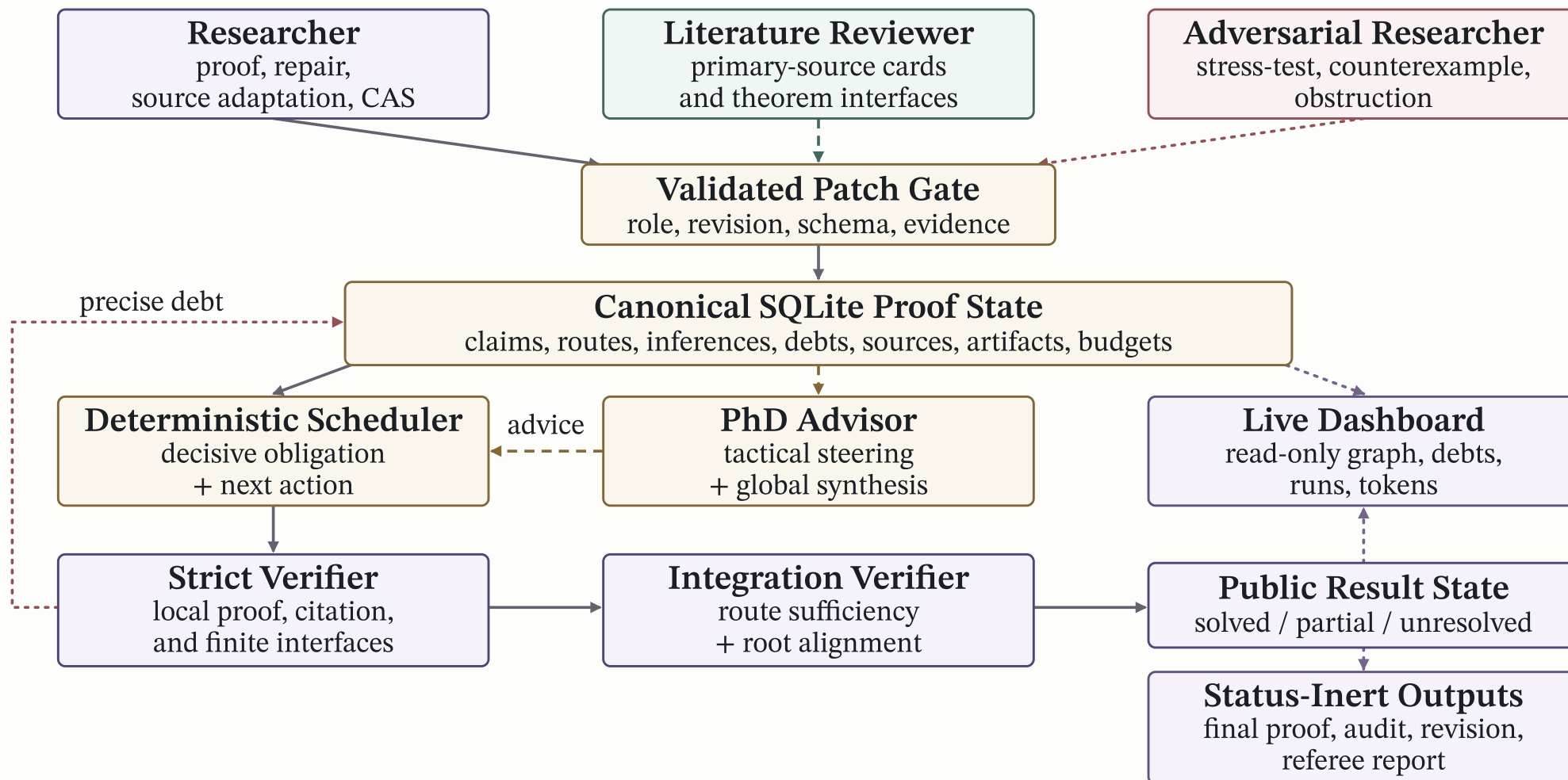
OPTIMAL BOUNDS IN BEND-AND-BREAK FOR FOLIATIONS

JIHAO LIU, ZEMING SUN, JIEDONG JIANG

Use of generative AI.

Rethlas obtained a complete proof.

Albilich



Redrawn from Figure 1 · Gong-Zeng-Yang (2026)

Albilich

Persistent proof state

An immutable root problem q_* .

A typed graph of **claims, routes, and inferences**.

Explicit **proof debts**: gaps, missing hypotheses, and unfinished obligations.

Tool calls

Theorem search: **TheoremSearch**.

Literature records and hypothesis checks.

SageMath · GAP · Macaulay2 · Singular

Versioned patches, computation transcripts, and a research history.

Albilich

Research

Researcher · adversarial
researcher · literature
researcher

Propose arguments,
counterexamples, and
computational evidence.



Verification

The **strict verifier** checks a
local claim.

The **integration verifier**
checks a route back to the
original problem.



Direction

The **PhD advisor** redirects
the search and chooses
routes.

Only a complete proof spine
reaches q_* .

Albilich

June 2026

We started experimenting with a research harness and obtained solutions to several open problems.



Expert review

Yong Yang checked the group-theoretic arguments with us.



Mathematical papers

We chose to release the group-theory results after verification and revision.

arXiv:1401.0300v47 [math.GR] 30 Sep 2026

Russian Academy of Sciences · Siberian Branch
SOBOLEV INSTITUTE OF MATHEMATICS

**UNSOLVED PROBLEMS
IN GROUP THEORY
THE KOUROVKA NOTEBOOK**

No. 21

Novosibirsk · 2026

Invariable generation

Independent conjugation is essential: the two conjugates need not use the same element of G .

Definition 2.1 (Gong–Yang–Zeng 2026)

Let G be a finite group. We say that G is invariably generated by elements $a, b \in G$ if

$$\langle gag^{-1}, hbh^{-1} \rangle = G \quad \text{for every } g, h \in G.$$

Dixon (1992); Kantor–Lubotzky–Shalev (2011)

Kourovka Problem 21.142

Fix distinct primes p and q .

Does every finite group embed into a finite group invariably generated by an element of order p and an element of order q ?

P. A. Zalesskii · The Kourovka Notebook

Kourovka Problem 21.142

Theorem 1.1 (Gong–Yang–Zeng 2026)

Let $p \neq q$ be fixed primes. Then there exists a finite group which does not embed into any finite group invariably generated by an element of order p and an element of order q .

Gong–Yang–Zeng, Alternating Groups and Embeddings into Groups Invariably Generated by Two Prime-Order Elements (2026)

Invariable generation

Lemma 2.2 (Gong–Yang–Zeng 2026)

Let G be a group and let $a, b \in G$. Then a and b invariably generate G if and only if there does not exist a proper subgroup $H < G$ and elements $g, h \in G$ such that $gag^{-1}, hbh^{-1} \in H$.

$$\langle gag^{-1}, hbh^{-1} \rangle \leq H < G.$$

The obstruction is a **single proper subgroup** meeting both conjugacy classes.

Kourovka Problem 21.142

Alternating sections

An embedding of A_m forces an alternating section in a composition factor S of the host group.

→

Prime-order conjugates

Analyze the induced action between S^t and $\text{Aut}(S) \wr \text{Sym}(t)$.

Use proper subgroups to exclude large alternating degree and large Lie rank.

→

A uniform bound

The remaining simple groups have bounded alternating sections.

Choose m larger than the bound depending on p, q .

Total 3-closure

Let $G \leq \text{Sym}(\Omega)$ act on a finite set Ω .

Definition

The **3-closure** $G^{(3),\Omega}$ consists of those $\sigma \in \text{Sym}(\Omega)$ such that, for every triple $(\alpha_1, \alpha_2, \alpha_3) \in \Omega^3$, some $g \in G$ agrees with σ on that triple.

The action is **3-closed** if $G^{(3),\Omega} = G$. The group is **totally 3-closed** if every faithful action on a finite set is 3-closed.

Wielandt (1964); Churikov–Praeger (2021)

Kourovka Problem 20.2

Are there any nonabelian simple groups of Lie type which are totally 3-closed?

$\mathrm{PSL}_2(7)$ is totally 3-closed.

The first witness led to a family-level classification.

Gong–Yang–Zeng (2026)

Total 3-closure

Theorem 1.1 (Gong–Yang–Zeng 2026)

Let $G = \text{PSL}_n(q)$ be a finite nonabelian simple group. Then the following hold.

1. If $n = 2$, then G is totally 3-closed if and only if $q \geq 7$ is a prime.
2. If $n = 3$, then G is totally 3-closed if and only if either $q = 3$, or q is a prime with $q \equiv 2 \pmod{3}$.
3. If $n = 4$, then G is not totally 3-closed for any q .
4. If $n \geq 5$ and $q > 2$, then G is not totally 3-closed.

Albilich

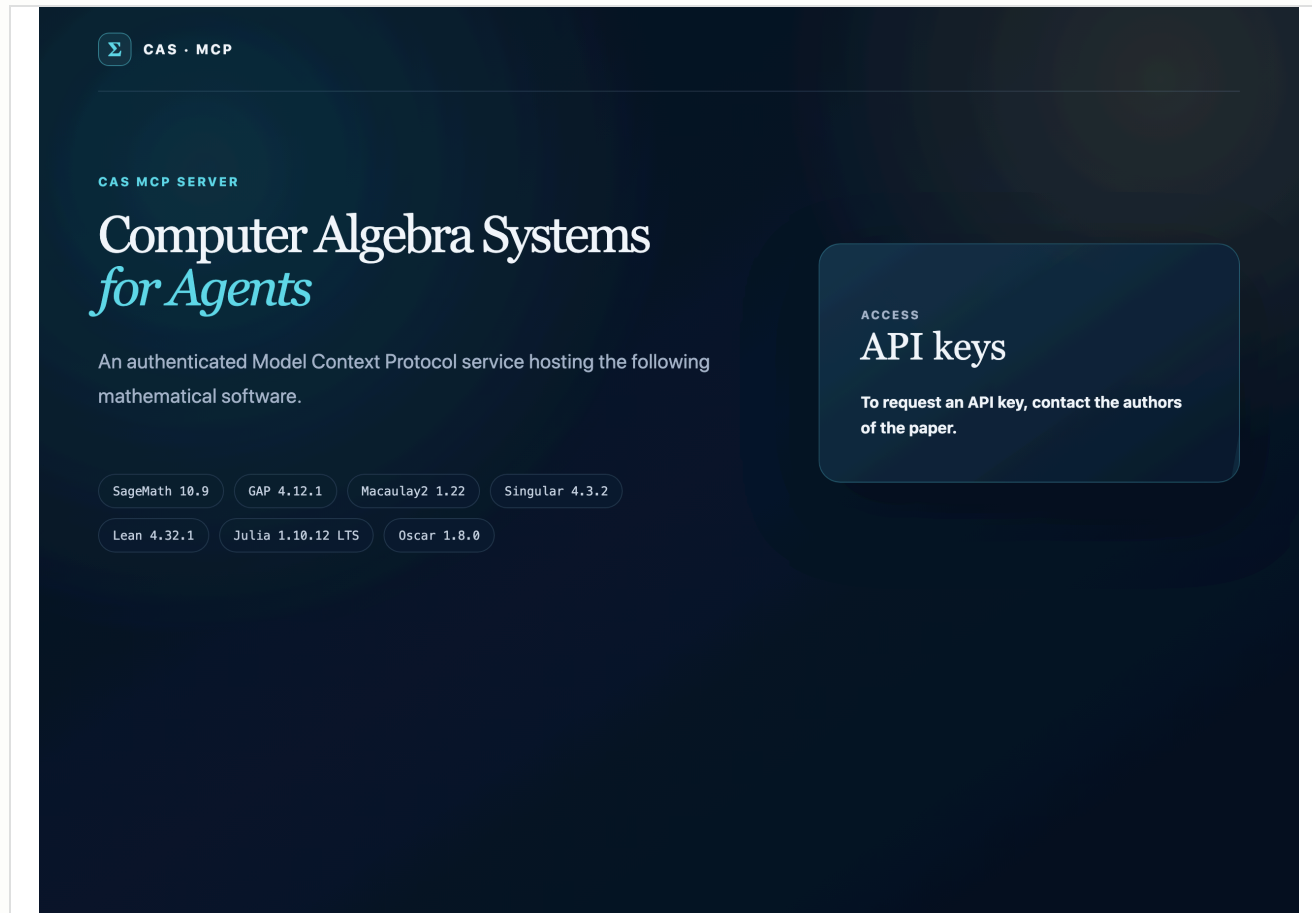
Kourovka 21.142 · same model and reasoning effort

	ADVISOR ENABLED	ADVISOR DISABLED
Verified claims	10	18
Routes concluding the root	1	0
Active compute	6.81 h	10.44 h
Gross tokens	29.7 M	54.9 M

The advisor-off run had more verified claims and no route concluding the root.

Gong–Zeng–Yang (2026), Table 3

Computer algebra systems for agents

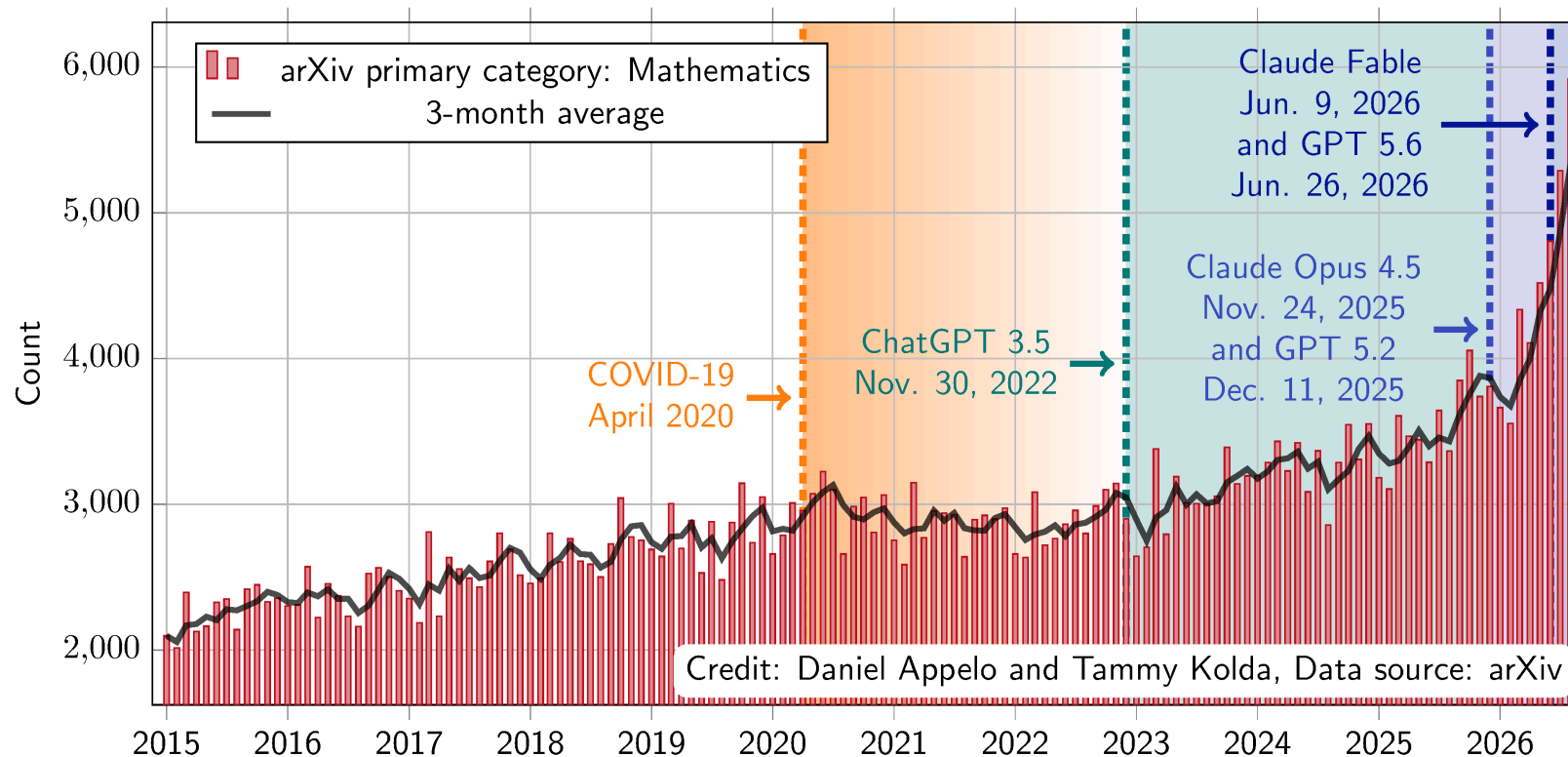


We released a **CAS MCP server** for agents to use mathematical software.

Open CAS MCP server

35.161.248.14

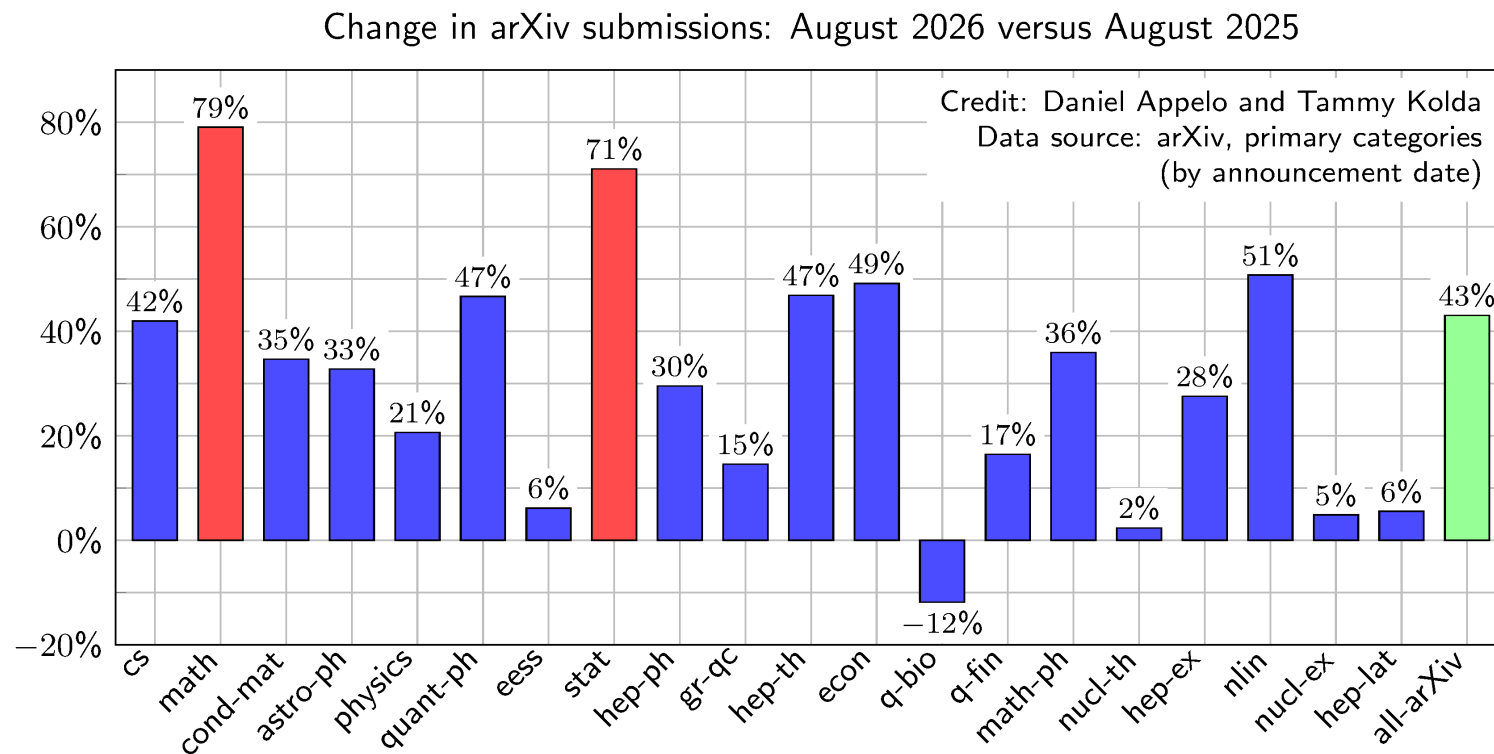
arXiv submissions



Mathematics · monthly submissions through August 2026.

Appelo–Kolda (September 2026), original plot

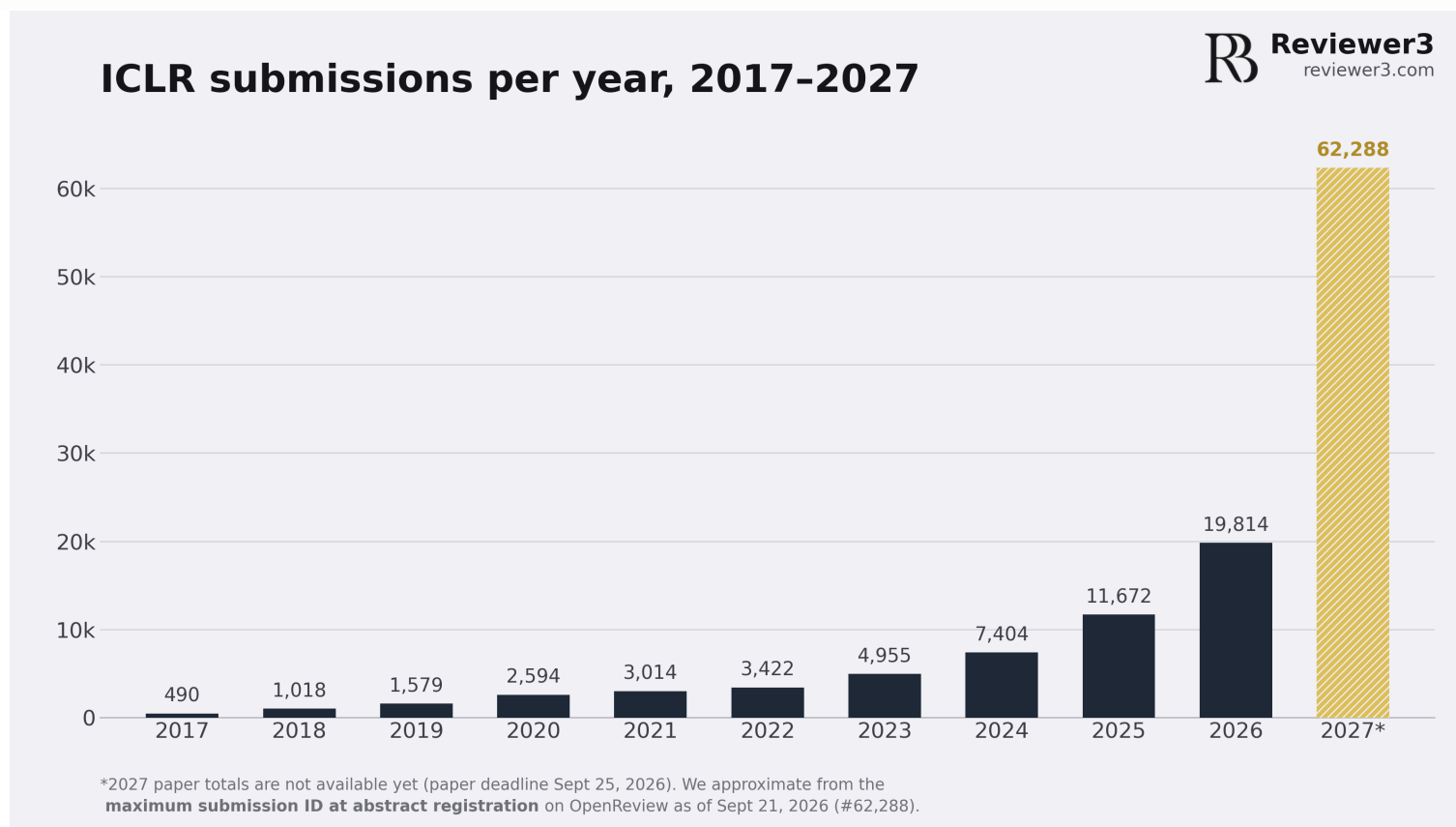
arXiv submissions



Math: 3,522 → 6,307 (+79%) **CS:** 10,779 → 15,309 (+42%)

Appelo–Kolda (September 2026) · August announcements, primary category.

ICLR 2027



62,288* · reported above all 2013–2026 editions combined.

*Highest abstract-registration ID, September 21; final paper count unconfirmed. [Source: Khalil \(2026\)](#)

Albilich: RSI metaharness

Recursive self-improvement

Use experience from proof searches to improve the research harness.

Learn from successful reductions, failed strategies, verifier feedback, and tool use.

Albilich v2 · UW Math AI Lab

Joint work with

Dean Light

Michael Theologitis

Evan Wang

PatternMiner

Conjecture generation

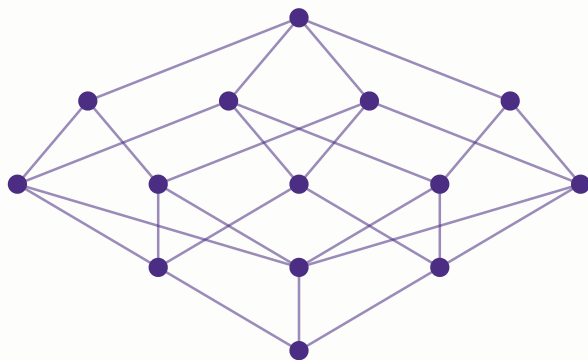
An LLM trains and interprets small models to propose mathematical statements.

Based on joint work with **Junaid Hasan**, **Xuanyu Yang**, **Emily Xia Meng**, and **Henry Kvinge**.

Submitted to ICLR 2027 main track

Combinatorial invariance conjecture

Bruhat interval $[1234, 3412]$



Hasse diagram · vertex labels forgotten

Poset alone?



Kazhdan–Lusztig polynomial

$$P_{1234,3412}(q) \\ = 1 + q$$

$$[x, y] \cong [x', y'] \text{ as posets} \implies P_{x,y}(q) = P_{x',y'}(q).$$

$$P_{x,y}(q) = \sum_{i \geq 0} \dim_{\mathbb{Q}} \mathcal{H}^{2i}(\mathrm{IC}_{X_y})_x q^i.$$

$X_y \subset \mathrm{Fl}(\mathbb{C}^n)$: Schubert variety, $x \leq y \in S_n$.

IC: rational intersection complex, unshifted on the smooth locus. [Kazhdan–Lusztig \(1980\)](#).

Davies et al. (2021)

For permutations $x \leq y$ in S_N , a graph neural network predicts coefficients of their Kazhdan–Lusztig polynomial $P_{x,y}(q)$ from the unlabelled Bruhat graph of $[x, y]$.

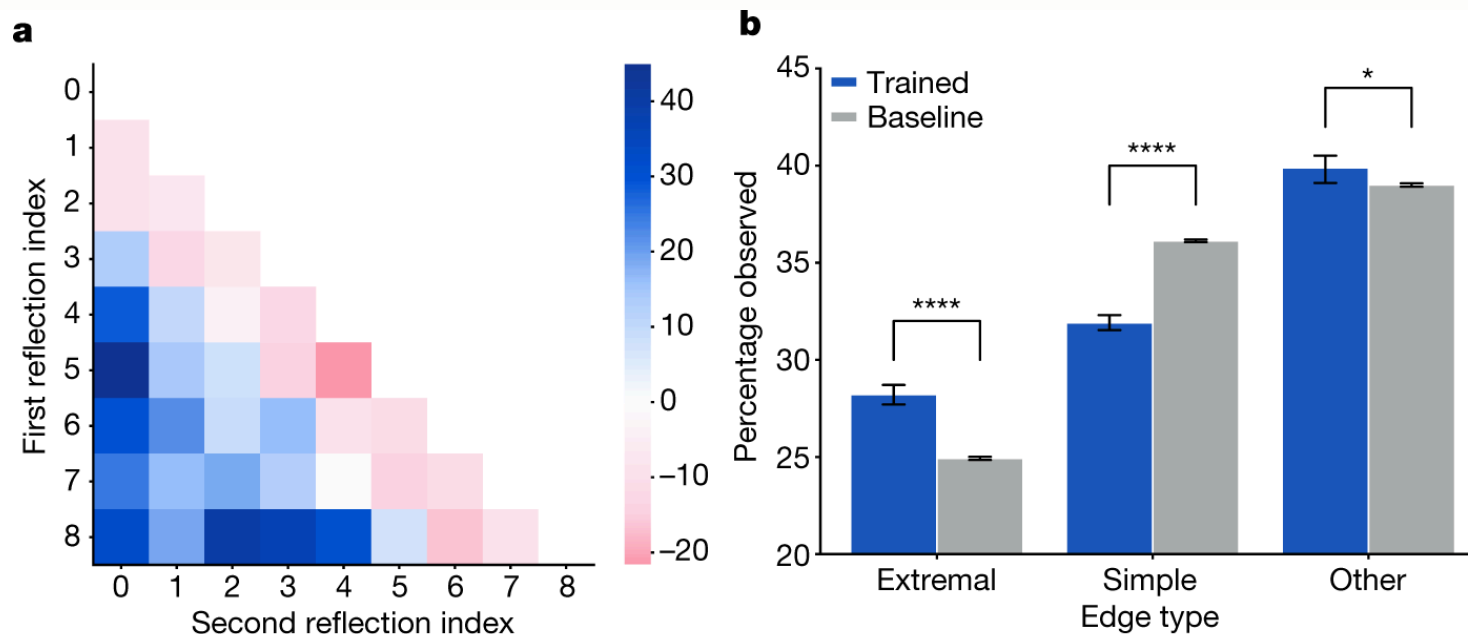


Figure 5(a,b). Saliency highlights **extremal reflections** $(0, i)$ and $(i, N - 1)$, although the network never sees these edge labels.

Davies et al. (2021)

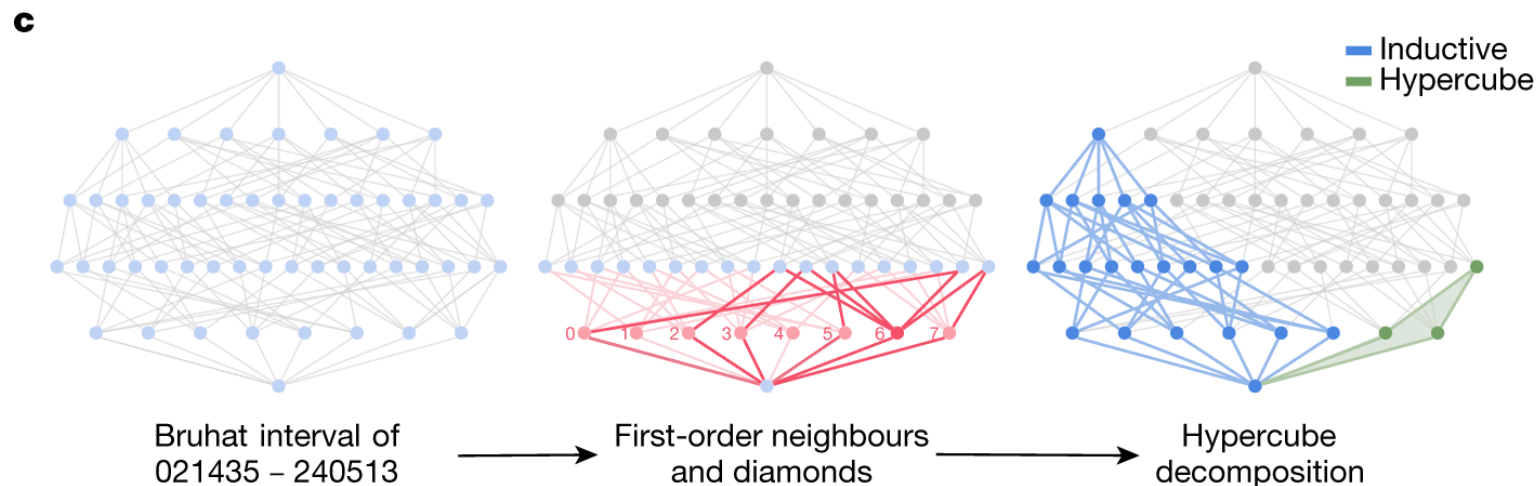


Figure 5(c). Blue: an interval in S_{N-1} . Green: a hypercube. The resulting formula recurses within $[x, y]$.

Theorem (Davies et al. 2021)

Every Bruhat interval admits a canonical hypercube decomposition along its extremal reflections, from which the KL polynomial is directly computable.

Proposed extension

The same computation for **any** hypercube decomposition would prove combinatorial invariance for symmetric groups. [Blundell et al. \(2022\)](#).

Interpreting small models

Huang et al. (2026)

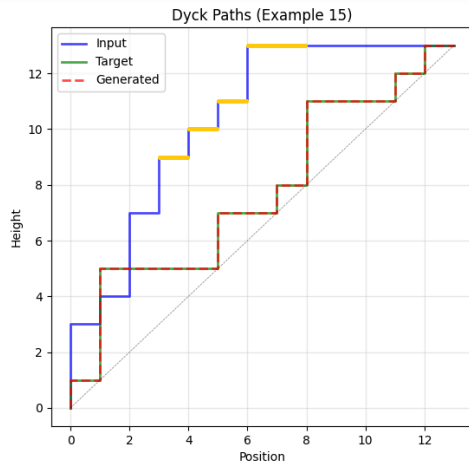


Figure 3(a). Attention and ablations led to a traversal algorithm proved equivalent to the zeta map.

He et al. (2025)

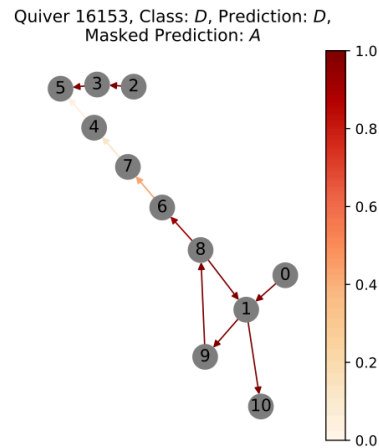
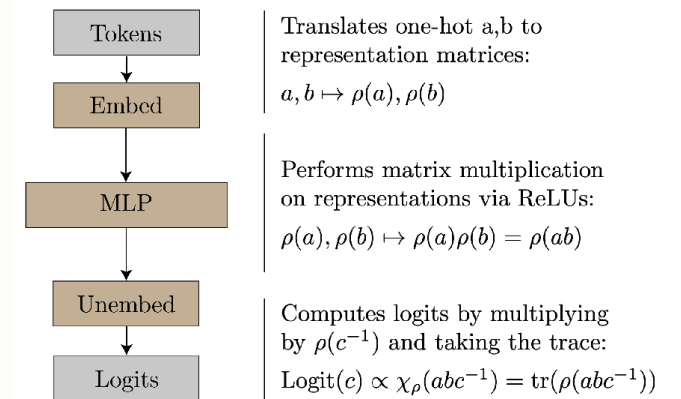


Figure 5, first example. Edge attribution identifies subgraphs used to recognize quiver mutation classes.

Nanda et al. (2023)

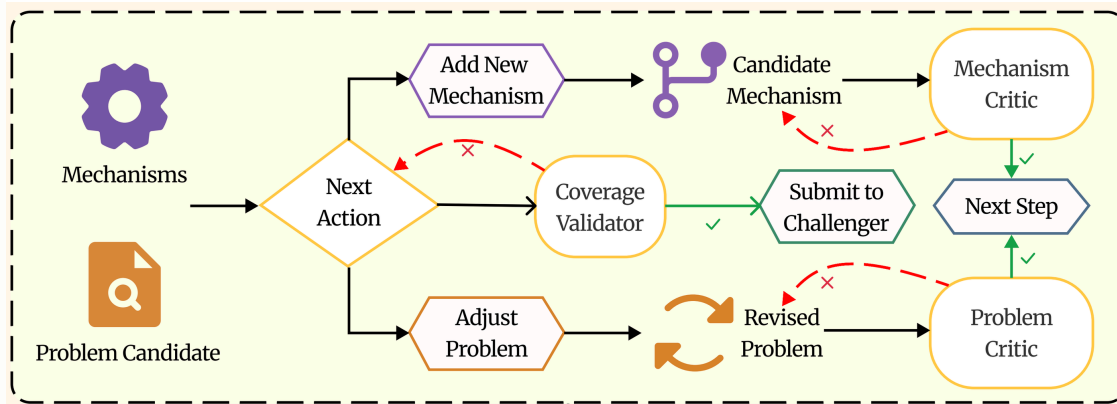
Chughtai et al. (2023)



Chughtai et al., Figure 1. Representation matrices explain group composition; Fourier features explain modular addition.

LLM conjecture generation

Long et al. (2026)



MECA · Figure 1, detail. Revise conjectures and supporting mathematical mechanisms using separate critics.

Kasaura et al. (2025)

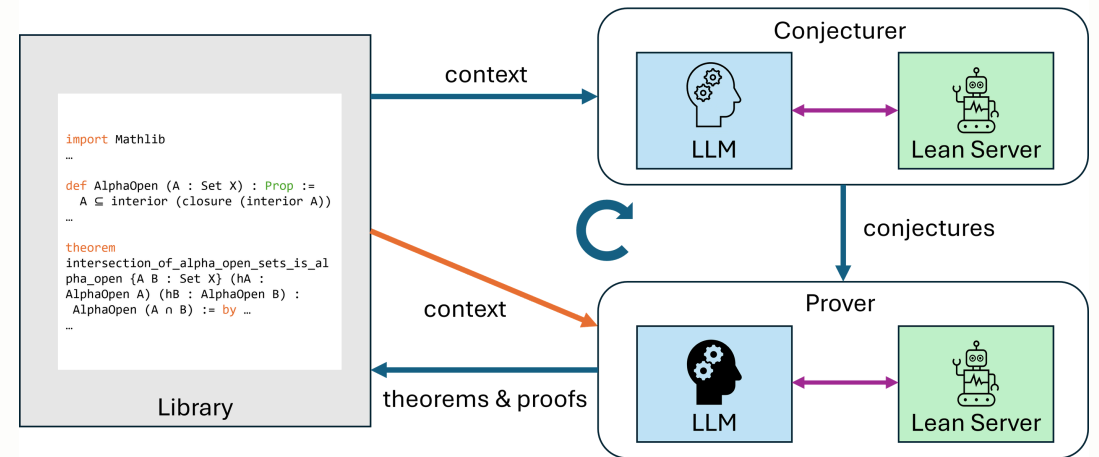


Figure 1. Verified conjectures and proofs return to a library that supplies the next prompts.

LLM conjecture generation

Onda et al. (2025)

```

8 theorem interior_eq_of_open : IsOpen s → interior s = s := by sorry
9
10 theorem new_theorem : IsOpen s → interior s = s := by exact?

```

▼MathlibDemo.lean:10:60

▼Tactic state

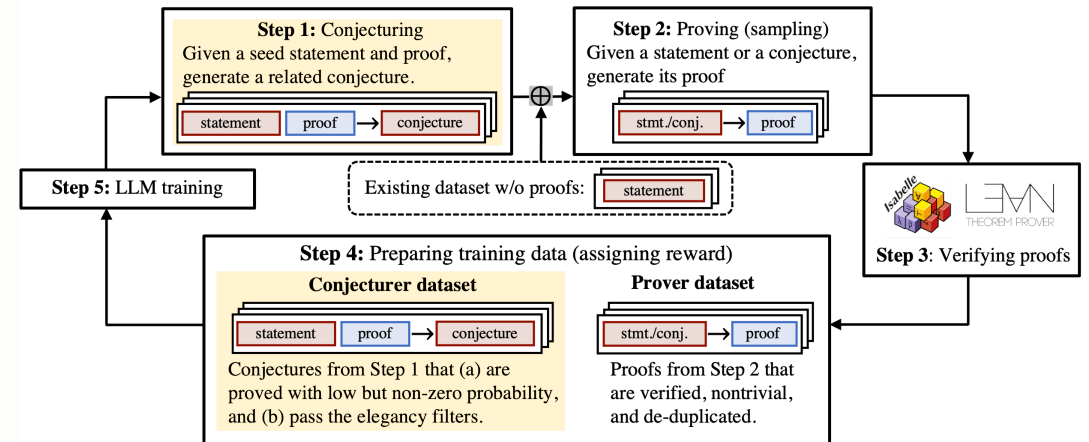
No goals

▼Suggestions

Try this: `exact fun a => interior_eq_of_open a`

LeanConjecturer · Figure 3, detail. Lean's `exact?` filters a statement already available in the context.

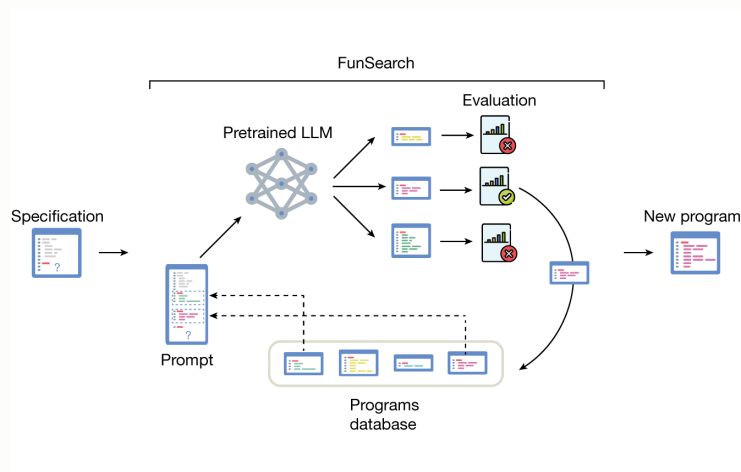
Dong–Ma (2025)



STP · Figure 1. Verified proofs train the prover; difficult but provable conjectures train the conjecturer.

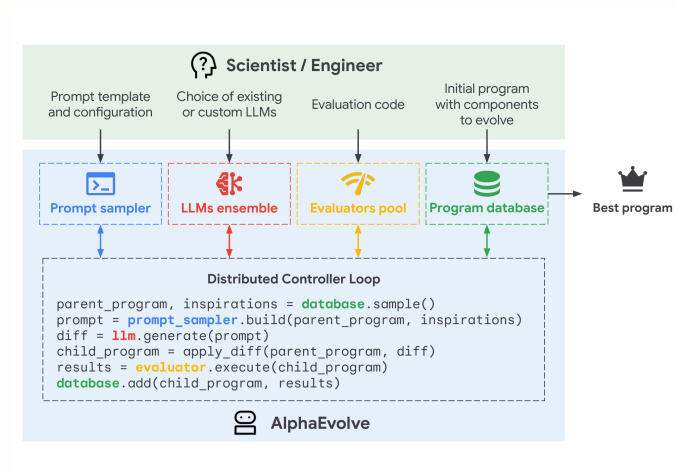
Searching for constructions

Romera-Paredes et al. (2024)



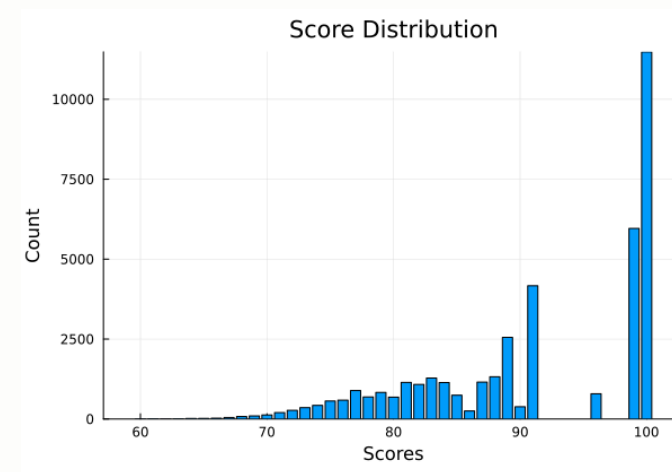
FunSearch · Figure 1. Evaluated programs return in later prompts.

Novikov et al. (2025)



AlphaEvolve · Figure 2. Evolve programs against a scientist-supplied evaluator.

Charton et al. (2024)



PatternBoost · Figure 4, final panel. Alternate local search and transformer training; scores are edge counts of triangle-free graphs on 20 vertices.

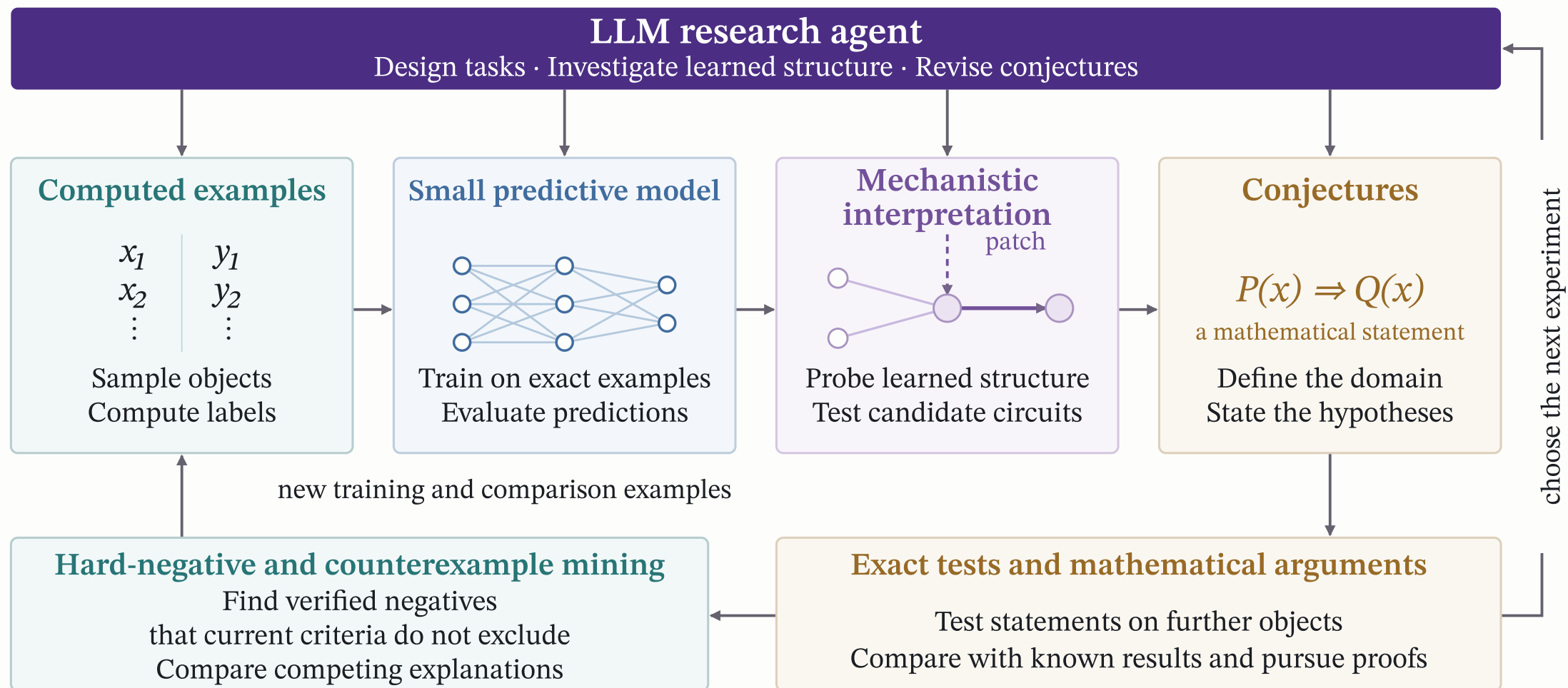
PatternMiner (2026)

APPROACH	SOURCE OF CONJECTURES	FEEDBACK CHANGES
Model interpretation	A researcher interprets learned structure	The conjecture and experiment
LLM conjecturing	The LLM reasons from literature, statements, and proofs	The statement, context, or prover
PatternMiner	An LLM trains and interprets small models on exact data	The conjecture and the next training set

Hard-negative mining closes the loop: examples satisfying the current necessary conditions but failing the target property are added with matching positives for retraining.

PatternMiner, §§2–3 · No controlled LLM-only comparison has yet been run.

PatternMiner



Redrawn from Figure 1 · PatternMiner (2026)

Positivity Problems and Conjectures in Algebraic Combinatorics

Richard P. Stanley¹
Department of Mathematics 2-375
Massachusetts Institute of Technology
Cambridge, MA 02139

version of 24 September 1999

1 Introduction.

Algebraic combinatorics is concerned with the interaction between combinatorics and such other branches of mathematics as commutative algebra, algebraic geometry, algebraic topology, and representation theory. Many of the major open problems of algebraic combinatorics are related to positivity questions, i.e., showing that certain integers are nonnegative. The significance of positivity to algebraic combinatorics stems from the fact that a nonnegative integer can have both a combinatorial and an algebraic interpretation. The archetypal algebraic interpretation of a nonnegative integer is as the dimension of a vector space. Thus to show that a certain integer m is nonnegative, it suffices to find a vector space V_m of dimension m . Similarly to show that $m \leq n$, it suffices to find an injective map $V_m \rightarrow V_n$ or surjective map $V_n \rightarrow V_m$. Of course the inequality $m \leq n$ is equivalent to the positivity statement $n - m \geq 0$, while the injectivity of the map $\varphi : V_m \rightarrow V_n$ is equivalent to the statement that $n - m = \dim \operatorname{coker}(\varphi)$ (where $\operatorname{coker}(\varphi)$ denotes the cokernel $V_n/\varphi(V_m)$ of φ). However, it is often more natural to deal with the inequality $m \leq n$ rather than with $n - m \geq 0$.

We will attempt here an overview of the outstanding open problems in algebraic combinatorics related to positivity. Naturally our choice is subjective, and we do not claim to be comprehensive.

¹Partially supported by NSF grant DMS-9500714

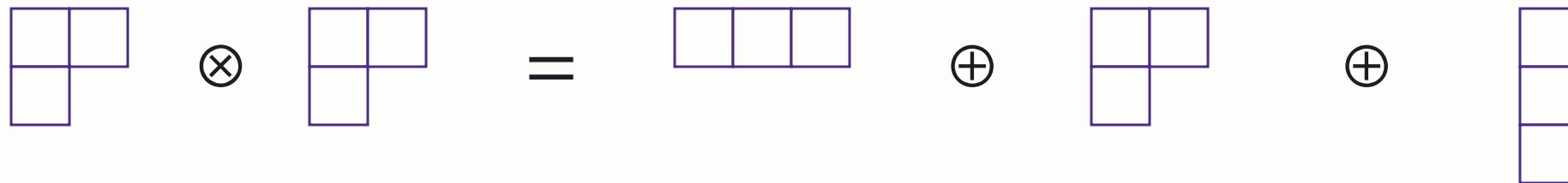
Kronecker coefficients

Partitions $\lambda \vdash n$ index irreducible representations V^λ of S_n .

$$V^\lambda \otimes V^\mu \cong \bigoplus_{\nu \vdash n} (V^\nu)^{\oplus g_{\lambda,\mu,\nu}}.$$

Stanley's Problem 10: find a combinatorial interpretation of $g_{\lambda,\mu,\nu}$.

For S_3 , we can see all three summands:



Each summand occurs once.

Kronecker coefficients

What the LLM trained

200 boosted trees, maximum depth 3.

Input: three partitions, encoded by their parts and padded with zeros, together with n .

Target: does $g_{\lambda,\mu,\nu} = 0$?

Exact labels from symmetric-group characters.

What the LLM tested

Training: **1,095 triples**, $3 \leq n \leq 7$.

Evaluation: **6,984 triples**, $n = 8, 9$.

MODEL	ACCURACY
Logistic regression	77.51%
Boosted trees	89.09%

PatternMiner, Table 9

Kronecker coefficients

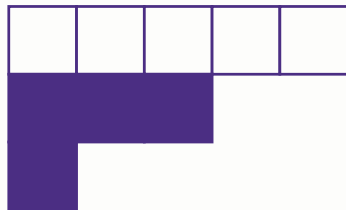
The LLM's analysis highlighted the **first parts**.

Theorem B.2.1 — Murnaghan's inequalities (Mishna–Rosas–Sundaram 2021, Eq. 22)

If $g_{\lambda, \mu, \nu} > 0$, then

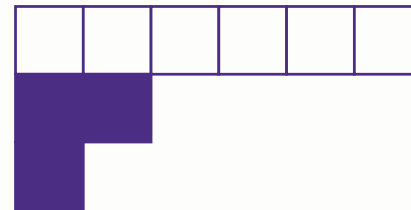
$$n - \lambda_1 \leq (n - \mu_1) + (n - \nu_1),$$

and the two inequalities obtained by permuting λ, μ, ν also hold.



$$\lambda = (5, 3, 1)$$

$$n - \lambda_1 = 4$$



$$\mu = (6, 2, 1)$$

$$n - \mu_1 = 3$$

The shaded boxes count the parts left after deleting the first row.

Kronecker coefficients

The LLM next mined zeros that satisfy the first-row inequalities. Their first two parts suggested deleting **two rows** instead.

For the following partitions of 3, the remaining sizes are 1, 0, 0.

$$g_{(1,1,1),(2,1),(2,1)} = 1, \quad 1 \not\leq 0 + 0.$$

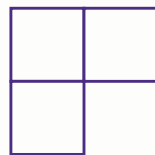
The proposed necessary condition fails.

$$\lambda = (1, 1, 1)$$



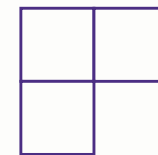
1 box left

$$\mu = (2, 1)$$



0 boxes left

$$\nu = (2, 1)$$



0 boxes left

Kronecker coefficients

Hooks and two-row partitions were frequent among the remaining predicted zeros. The LLM trained the same kind of trees on $g \leq 1$ and $g \leq 2$.

Theorem B.2.2 (Rosas 2001, Corollaries 15 and 17)

For $\lambda, \mu, \nu \vdash n$, the following bounds hold, with the third partition arbitrary:

$$g_{\lambda, \mu, \nu} \leq 2 \quad \text{if two partitions are hooks,}$$

$$g_{\lambda, \mu, \nu} \leq 3 \quad \text{if two partitions are a hook and a two-row partition.}$$

A hook has shape $(n - r, 1^r)$.

The later classifier respected the two-hook bound on all 5,236 applicable test triples at $n = 10, 11$.

Schubert structure constants

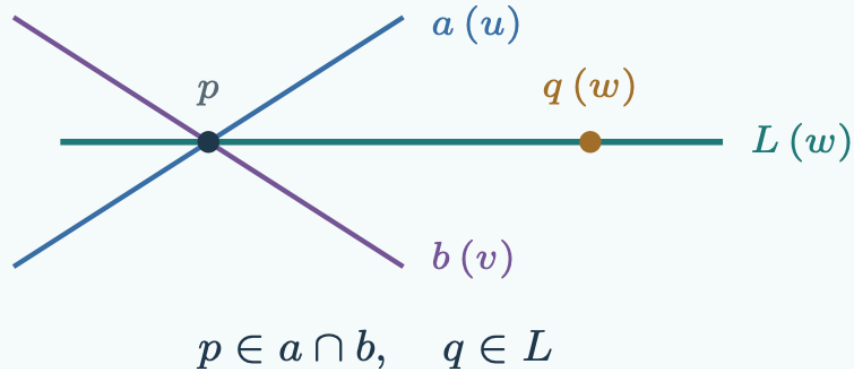
On the complete flag variety $\mathrm{Fl}_n(\mathbb{C})$, let σ_u be the Schubert class indexed by $u \in S_n$.

$$\sigma_u \sigma_v = \sum_w c_{u,v}^w \sigma_w, \quad c_{u,v}^w \in \mathbb{Z}_{\geq 0}.$$

We ask **when** $c_{u,v}^w > 0$: do the incidence conditions have a common flag?

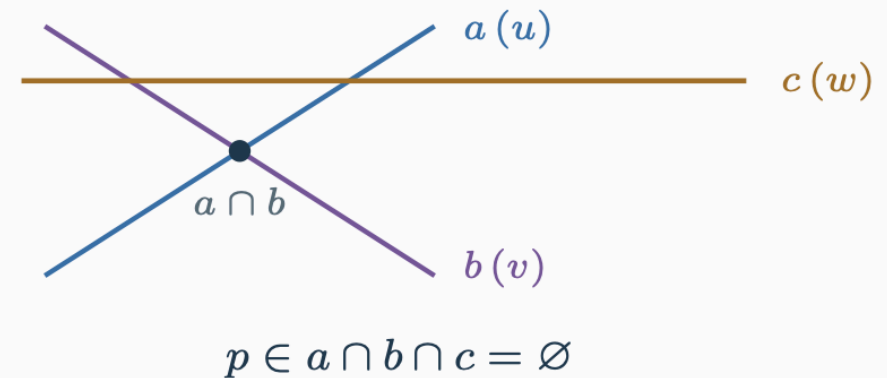
One flag

$$c_{213,213}^{312} = 1$$



No flag

$$c_{213,213}^{231} = 0$$



Schubert structure constants

The LLM trained a width-192 transformer on **1,272,767 exact triples** to predict $c_{u,v}^w > 0$. Its predictions suggested the descent condition.

For a permutation x , $\text{Des}(x) = \{i : x(i) > x(i+1)\}$.

Proposition B.1.1 (Knutson 2003)

If $c_{u,v}^w > 0$, then $\text{Des}(w) \subseteq \text{Des}(u) \cup \text{Des}(v)$.

IN THE TWO EXAMPLES	$\text{Des}(U) \cup \text{Des}(V)$	$\text{Des}(W)$
$c_{213,213}^{312} = 1$	$\{1\}$	$\{1\}$
$c_{213,213}^{231} = 0$	$\{1\}$	$\{2\}$

Schubert structure constants

The LLM retrained the transformer on triples passing the earlier conditions, then searched **rank-matrix features**.

$$r_x(i, j) = \#\{a \leq i : x(a) \leq j\}.$$

$\text{Ess}(w)$ is the set of southeast corners of the Rothe diagram $D(w) = \{(i, j) : j < w(i), i < w^{-1}(j)\}$.

Proposition B.1.2 (Fulton 1997)

If $(i, j) \in \text{Ess}(w)$ and $r_u(i, j) = r_v(i, j) = r_w(i, j)$, then $c_{u,v}^w = 0$.

This removed **133 of 45,000** remaining zeros, with **0 violations among 59,979** nonzero coefficients.

Schubert structure constants

The LLM mined zeros that survive **projection tests**: retain selected subspaces of a flag and test whether the projected conditions have a common solution.

An exact example in S_5 :

$$u = 14235, \quad v = 24153, \quad w = 45123, \quad c_{u,v}^w = 0.$$

Here $\text{Des}(u) \cup \text{Des}(v) = \{2, 4\}$.

Retain dimension 2

$$V_2 \subset \mathbb{C}^5$$

Nonempty

Retain dimension 4

$$V_4 \subset \mathbb{C}^5$$

Nonempty

Retain both

$$V_2 \subset V_4 \subset \mathbb{C}^5$$

Empty

Schubert structure constants

The LLM kept finding examples requiring more retained dimensions. We redirected it toward **a construction valid at every size**.

Here $\ell(x)$ counts inversions of x .

Theorem B.1.3 (PatternMiner)

For every $n \geq 5$, there exist $u, v, w \in S_n$ with $\ell(w) = \ell(u) + \ell(v)$, $c_{u,v}^w = 0$, and

$$\text{Des}(w) \subseteq D = \text{Des}(u) \cup \text{Des}(v), \quad |D| = n - 3,$$

such that every projection retaining a proper subset $E \subsetneq D$ has a nonempty generic intersection.

Schubert structure constants

The LLM next computed **exact equivariant products under left descents**.

For double Schubert polynomials $\mathfrak{S}_u(x; t)$,

$$\mathfrak{S}_u(x; t) \mathfrak{S}_v(x; t) = \sum_w c_{u,v}^w(t) \mathfrak{S}_w(x; t).$$

Write $\alpha_i = t_{i+1} - t_i$. Graham positivity gives

$$c_{u,v}^w(t) \in \mathbb{Z}_{\geq 0}[\alpha_1, \dots, \alpha_{n-1}].$$

Setting $t = 0$ recovers the ordinary coefficients $c_{u,v}^w$.

Graham (2001); PatternMiner, §B.1.6

Schubert structure constants

s_i exchanges the values $i, i + 1$; $\leq$ denotes Bruhat order. The positive roots are

$$\Phi^+ = \{\alpha_a + \cdots + \alpha_b : 1 \leq a \leq b < n\}.$$

Theorem B.1.5 (PatternMiner)

For $u, v, w \in S_n$ with $s_i u < u \leq w$ in Bruhat order, the difference

$$c_{u,v}^w(t) - \alpha_i c_{s_i u, v}^w(t)$$

is a nonnegative integer combination of products of distinct roots in $\Phi^+ \setminus \{\alpha_i\}$.

This proves Conjectures A and B of [Chen et al. \(2026\)](#) on equivariant nonvanishing.

Monomial immanants

For an $n \times n$ matrix M and a class function ϕ on S_n , define

$$\text{Imm}_\phi(M) = \sum_{w \in S_n} \phi(w) \prod_{i=1}^n M_{i,w(i)}.$$

For a skew shape $D = \mu/\nu$, use its Jacobi–Trudi matrix

$$H_D = (h_{\mu_i - \nu_j - i + j})_{i,j=1}^n.$$

Here h_r is the complete symmetric function, $h_0 = 1$, and $h_r = 0$ for $r < 0$. Choose ϕ^λ with Frobenius characteristic m_λ .

Monomial immanants

For a content vector $\mathbf{c}$ of total size $N = |D| \geq 4$, let r count its positive entries and t its entries at least two. The Kostka number $K_{\theta, \mathbf{c}}$ counts semistandard tableaux of shape θ and content $\mathbf{c}$.

For the first non-hook shape, the LLM obtained

$$K_{(N-2,2), \mathbf{c}} = \binom{r-1}{2} + t - 1.$$

The hook contribution is $\binom{r-1}{2}$. The LLM tested the correction $t - 1$, found a negative decomposition coefficient, and proved that the hook contribution cancels it.

This suggested extending the argument to $\theta_2 \leq 2$.

Monomial immanants

The hook case follows from **Lesnevich (2024)** and **Hikita (2025)**. The result extends it to arbitrarily many rows of length two below the first row.

Theorem B.5.2 — Schur-positivity consequence (PatternMiner)

Let $D = \mu/\nu$ be a nonempty skew shape of size N , represented with n rows. For every $\theta \vdash N$ with $\theta_2 \leq 2$ and every $\lambda \vdash n$,

$$[s_\theta] \operatorname{Imm}_{\phi^\lambda}(H_D) \geq 0.$$

Here $[s_\theta]$ denotes the coefficient of the Schur function s_θ , and a missing second part is zero.

The proof gives a nonnegative Stanley–Stembridge character decomposition.

PatternMiner

Character-table row sums

The LLM trained a linear model to pair negative tableaux with positive tableaux. Held-out collisions refuted injectivity.

For a fixed tail $\nu \vdash k$, the LLM derived an exact evaluation formula for the row indexed by $(n - k, \nu)$.

Theorem B.3.2; Table 16

Kazhdan–Lusztig coefficients

The LLM’s best decision stumps classified **30/33** recurrence terms, but failed on a withheld pair.

The LLM proved restrictions on a possible failure of combinatorial invariance at interval length **13**.

Theorem B.4.5; Appendix B.4.1

CircuitPilot

Learning Circuit Optimization through Imitation and Self-Play

Based on joint work with **Weikun K. Zhang, Rohan Pandey, Bhaumik Mehta, Kaijie Jin, Naomi Morato, Archit Ganapule, and Jarod Alper.**

CircuitBuilder · ICLR 2026 Workshop on AI with Recursive Self-Improvement

FactorLibrary · 3rd AI for Math Workshop, ICML 2026

CircuitPilot · Submitted to ICLR 2027 main track

Learning to search for circuits

AlphaZero

Policy and value learning alternate with tree search and self-play.

Silver et al. (2018)

AlphaTensor-Quantum

Reduce quantum T -gate counts through learned tensor decompositions.

Ruiz et al. (2024)

AlphaTensor

Discover matrix-multiplication algorithms by playing a tensor-decomposition game.

Fawzi et al. (2022)

Quasar

Equality saturation searches equivalent quantum circuits within a rewrite bound.

Yang–Raun–Tao–Gu (2026)

Polynomial circuits

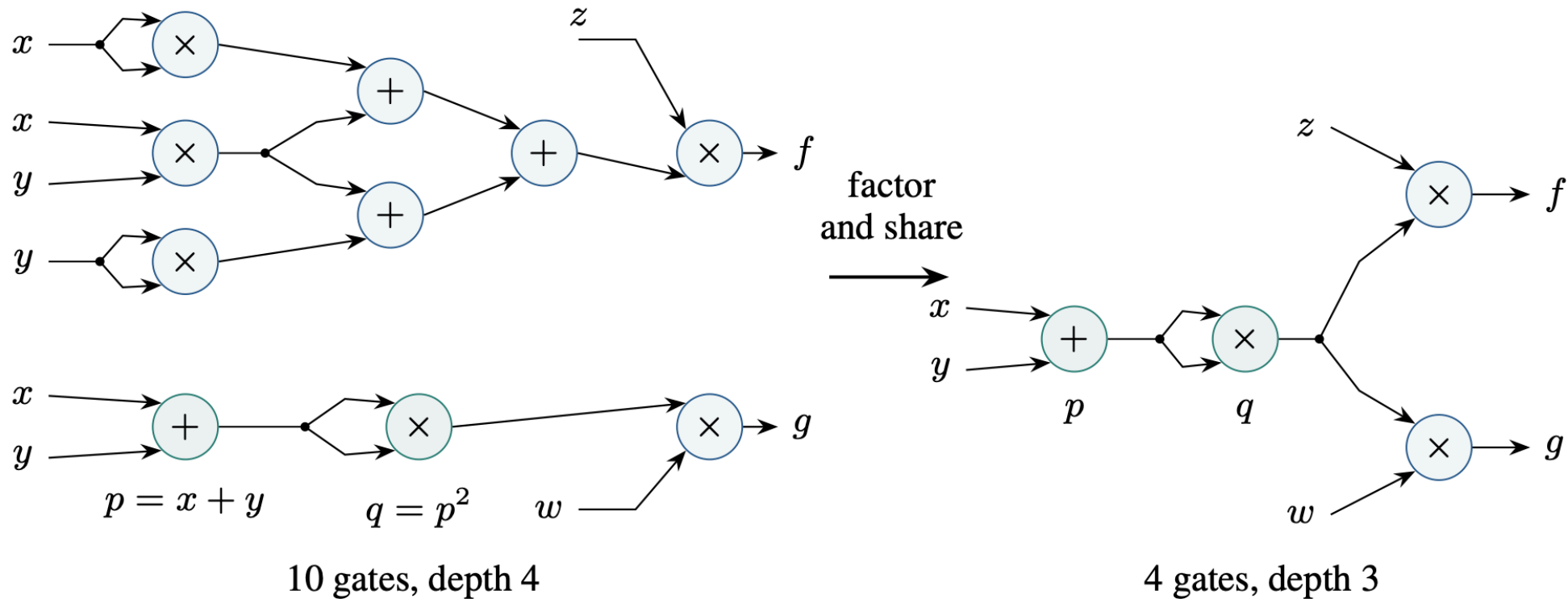


Figure 1. Inputs x, y, z, w produce outputs f, g . Factoring lets both outputs reuse the squared sum: **ten gates become four.**

Boolean circuits

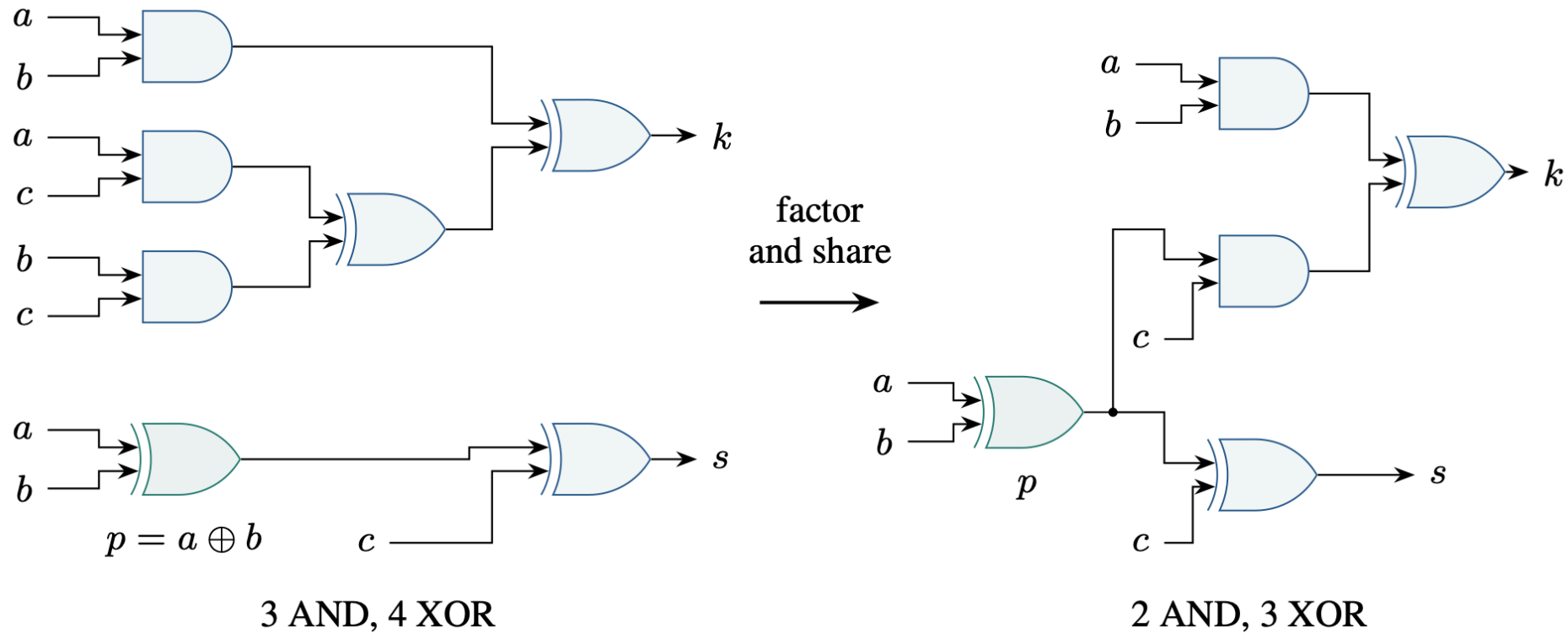


Figure 2. A full adder takes bits a , b , c and returns sum s and carry k . Sharing the highlighted XOR saves **one AND and one XOR**.

Quantum circuits

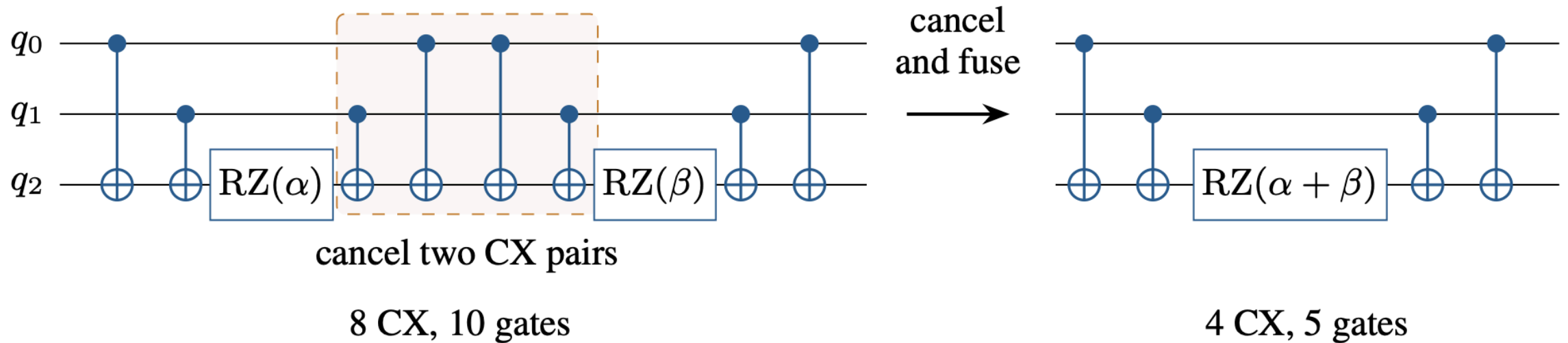


Figure 3. The wires carry three qubits; α, β are rotation angles. Cancel matching controlled-X (CX) gates, then combine the adjacent rotations.

Circuit synthesis and optimization

CircuitBuilder

Construct a circuit for a target polynomial, one gate at a time.

ICLR 2026 Workshop on AI with Recursive Self-Improvement

FactorLibrary

Reuse previously constructed subexpressions as subgoals.

3rd AI for Math Workshop · ICML 2026

CircuitPilot

Start with a circuit. Learn which optimization procedure to run next.

Submitted to ICLR 2027 main track

CircuitPilot

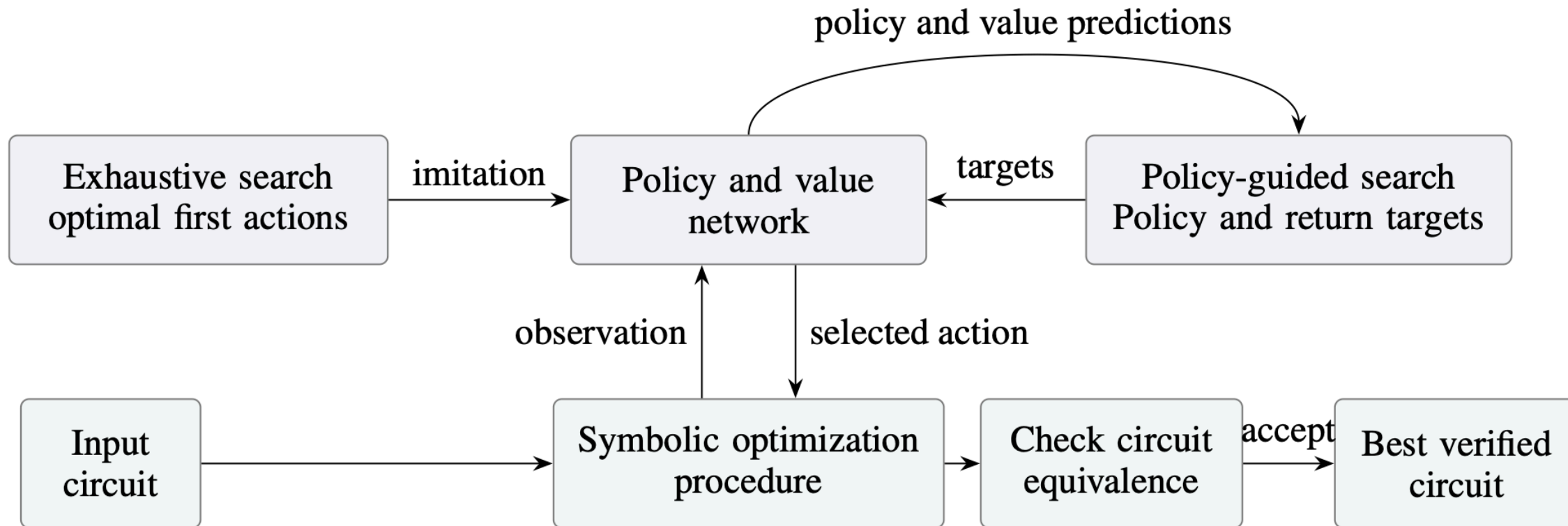


Figure 4. Imitate exhaustive short searches; improve through self-play. At deployment, select procedures directly and check circuit equivalence before accepting a result.

Quantum circuit optimization

70 wins · 82 ties · 9 losses on 161 three-qubit circuits.

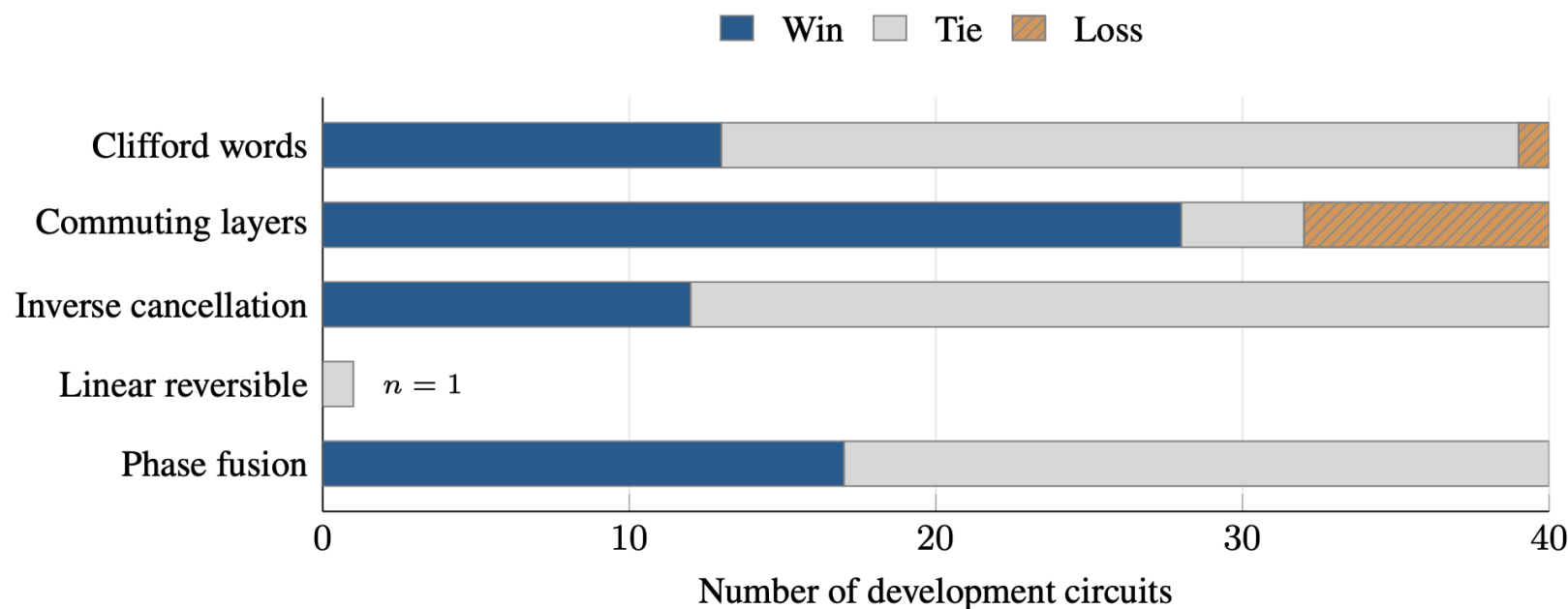


Figure 5. Compared with step-8 Quasar; time limits are 30 minutes for CircuitPilot and 1 minute for Quasar. Wins mostly reduce gate count or depth at fixed CX count.

Quantum circuit optimization

REWRITE RULES	WINS	TIES	LOSSES
Initial rules	46	81	34
Remove an expansion rule	57	91	13
Add simplification rules	70	82	9

Even a valid rewrite can hurt when search has a budget.

Table 2. Same 161 development circuits and step-8 Quasar reference.

Polynomial and Boolean experiments

Polynomial optimization

400 development circuits

An imitation-only policy uses **0.25% fewer operations** than the best fixed procedure sequence.

Running all three fixed sequences and keeping the best result for each input still does better.

Boolean input coverage

98 of 100 benchmarks

The environment loads and encodes the IWLS 2026 circuits as model inputs.

This study evaluates encoding; Boolean optimization performance remains to be measured.

CircuitPilot

Start with a circuit. Follow the optimization choices. Compare the resulting circuit and its cost.

[Open demonstration \(sign-in required\)](#)

CircuitPilot demonstration

AXIOM MATH · OCTOBER 6, 2026

JMM 2027 & UW Math AI Lab

Small models, mathematical datasets, and learned search

Based on joint work with **Junaid Hasan**, **Henry Kvinge**, and the **UW Math AI Lab community**.

AMS Special Session · JMM 2027 · UW Math AI Lab

AI for Mathematics at Small Scale

Mathematical Discovery with User-Controlled Models and Data

AMS Special Session · JMM 2027

Junaid Hasan · Henry Kvinge · Michael Ruofan Zeng

Friday, January 15 · Chicago

Afternoon session: 1–5 pm · McCormick Place, S-501a



Session program

Small models · interpretability · learned search

UW Math AI Lab



Spring Demo Day · June 8, 2026

Research together

Faculty, graduate mentors, and
undergraduate researchers.

Dhruv Bhatia & Michael R. Zeng

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Large and small models for mathematical discovery

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